

## **Automatic Grouping of Real and Fake News Stories by Topic**

### **Context/Introduction to problem:**

The spread of false information in news stories online has become an increasingly prevalent problem. Studies have shown that humans are only successful in identifying fake news stories around 70 percent of the time [1]. Given the abundance of news stories we encounter every day, this relatively low accuracy makes the average person susceptible to a great deal of false information.

It is a reasonable hypothesis to make that fake and real news stories will have qualitatively different writing styles. For example, a fake news story will likely be written very hyperbolically, or with a more extreme position taken in the article itself. Fake news stories may also not be written as carefully, and may have more grammatical mistakes. Quantifying these sorts of patterns is in general a very complicated task, but finding insight into them is crucial for understanding fake news and preventing the spread of misinformation.

Additionally, the writing style may also vary by topic. A story about geopolitics will likely be written in a different way than a story about a local election. Given that the writing style of a news story may vary by both reliability and topic, it is beneficial to find a way to decouple these two variables. If one can successfully find a way to split news stories up by topic, we can independently analyze how real versus fake stories are written about that topic. By isolating stories by the topic they are written about, we can more reliably determine the differences between real and fake news stories of differing topics.

Investigating the properties of fake news, as well as reliably being able to determine the topic of a large number of news stories [3], is paramount for protecting democratic institutions, and ensuring that people are able to access true and reliable information.

### **Project Deliverable:**

In this project, you are a data scientist tasked with taking a dataset of real and fake news stories, and trying to determine the differences between real and fake news stories of different topics. The problem is, the news stories you have aren't split into different topics by default, and because of the size of the dataset, you could not possibly manually classify them all. You are tasked with determining how to split news stories by topic. To do this, you will use two approaches, first you will convert the text-based data into numerical data using TF-IDF vectorization, and then you will classify the numerical clusters using K-means clustering [2].

From this modeling approach, you will try to gain insight into the qualitative and quantitative differences between real and fake news stories, as a function of the article topic, with the broader goal of developing better ways to combat fake news.

[1] *Fake news detector algorithm works better than a human*. Michigan News.  
<https://news.umich.edu/fake-news-detector-algorithm-works-better-than-a-human/>

[2] *Clustering text documents using K-means*. scikit. (n.d.).  
[https://scikit-learn.org/stable/auto\\_examples/text/plot\\_document\\_clustering.html](https://scikit-learn.org/stable/auto_examples/text/plot_document_clustering.html)

[3] Shahane, S. (2023). *Fake news classification* [Data set]. Kaggle.  
<https://www.kaggle.com/datasets/saurabhshahane/fake-news-classification>